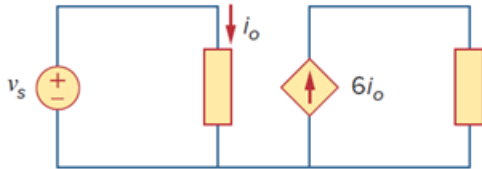


Problem

The dependent source in Fig. 1.22 is:

- (a) voltage-controlled current source
- (b) voltage-controlled voltage source
- (c) current-controlled voltage source
- (d) current-controlled current source

Figure 1.22:



Step-by-step solution

Step 1 of 3

Refer to Figure 1.22 in the textbook.

Consider the following circuit diagram as shown in Figure 1.

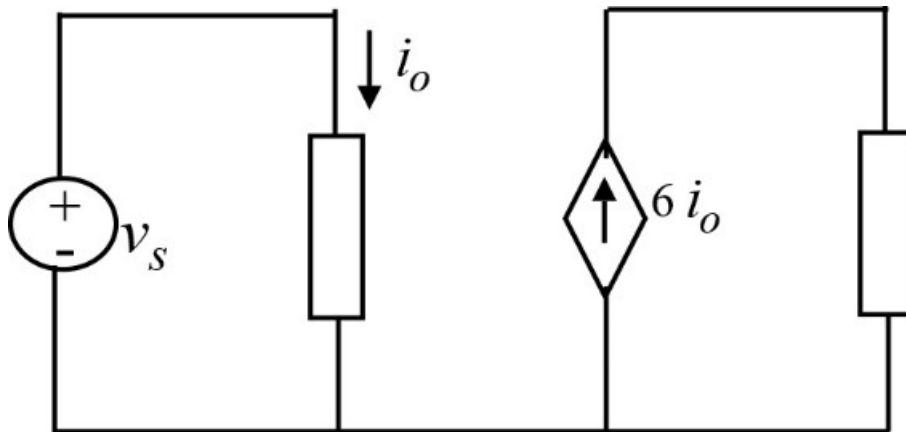


Figure 1

Step 2 of 3

An ideal dependent (or controlled) source is an active element in which the source quantity is controlled by another voltage or current.

Dependent sources are usually designated by diamond-shaped symbols.

Therefore, option (a), (b), and (c) are incorrect.

Step 3 of 3

Since the diamond-shaped symbol in Figure 1 is a current source.

Thus, the dependent source shown in Figure 1, is a current-controlled current source (CCCS).

Hence, the correct answer is option (d).